

1. Which statement about uniform circular motion is correct?

A. speed constant B. velocity constant C. net force points to center D. KE changes

Answer: C. The net force points to center.

2. A 0.5kg ball moves in a horizontal circle of radius 0.4m at 2 rad/s. Find the force.

Answer: $F = m * \omega^2 * r = 0.5 * 4 * 0.4 = 0.8 \text{ N}$.

3. A rod of length L cuts a magnetic field B with velocity v. Find induced emf.

Answer: By Faraday law, $E = B * L * v$.